Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18RADE552]

FIFTH SEMESTER B.Tech. EXAMINATION, DECEMBER–2021

INTELLIGENT CONTROL

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Derive an artificial neuron from a biological neuron. Explain the mathematical model of an artificial neuron. **07**

b. Distinguish Supervised and Unsupervised learning methods with proper diagrams. **07**

2. a. Explain the feedback networks in artificial neural networks. **07**

b. Describe the Discrete Time Hopfield Net with a suitable example. **07**

3. a. Let the three fuzzy sets have been given as follows: **07**

$$\underset{\sim}{A} = \left\{\frac{0.3}{1} + \frac{0.7}{2} + \frac{0.4}{3} + \frac{0.1}{4}\right\}$$

$$\underset{\sim}{B} = \left\{\frac{0.2}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.3}{4}\right\}$$

$$\underset{\sim}{C} = \left\{\frac{0.6}{1} + \frac{0.1}{2} + \frac{0.2}{3} + \frac{0.4}{4}\right\}$$

Find

i. $\underset{\sim}{A} \cap \underset{\sim}{B}$

ii. Prove Distributivity Property

b. Let the elements of the Universe be **07**

$X = \{x_1, x_2\}$, $Y = \{y_1, y_2\}$, and $Z = \{z_1, z_2, z_3\}$

The relations between them are given below

$$\underset{\sim}{R} = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.7 & 0.5 \\ 0.8 & 0.4 \end{bmatrix} \end{array} \qquad \underset{\sim}{S} = \begin{array}{c} \\ y_1 \\ y_2 \end{array} \begin{array}{c} \begin{array}{ccc} z_1 & z_2 & z_3 \end{array} \\ \begin{bmatrix} 0.9 & 0.6 & 0.2 \\ 0.1 & 0.7 & 0.5 \end{bmatrix} \end{array}$$

Find the Composition matrix using max min methods.

OR

4. a. Explain the following defuzzification Methods. **07**

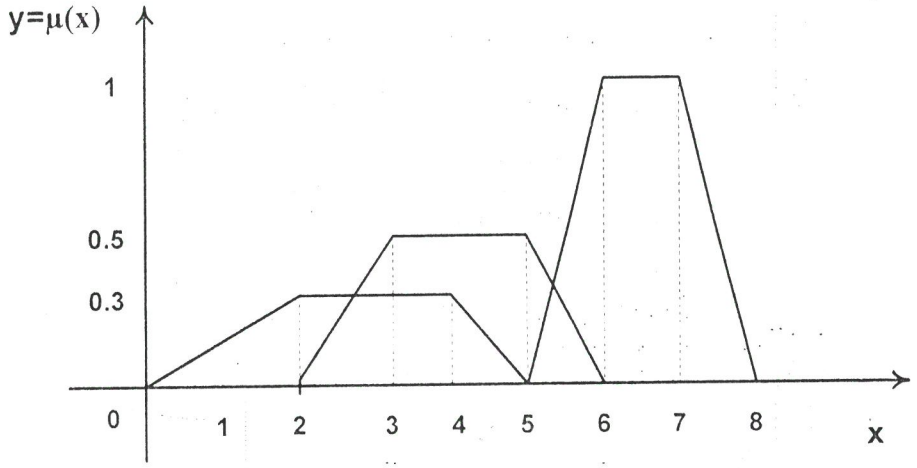
i. Mean Maximum Method

ii. Centre of Sums Method

b. Obtain the defuzzified value of the following diagram by using **07**

i. Maximum Membership Principle

ii. Centroid Method



5. a. Discuss the membership assignment methods process in fuzzy logic applications. **07**

b. Develop a smart home heating system using Intelligent controllers. **07**

OR

6. a. Design an Aircraft Landing System by using Fuzzy Logic Controller. **07**

b. Construct a Fuzzy logic controller for a Motor speed control. **07**

7. a. Explain the various steps involved in Genetic Algorithm. **07**

b. Find **07**

i. The offspring after crossover for the population 11000 and 01101 at crossover point of 3.

ii. The offspring after crossover for the population 11000 and 10011 at crossover point of 4.

iii. The offspring after mutation of 01100 at level 3.

OR

8. a. Elaborate on utilization of Tabu Search Technique in the optimization Problems. **07**

b. Demonstrate an algorithm for Travelling Salesman Problem using Ant Colony optimization. **07**